# Advanced Pandas

## Assignment

PW SKILLS

**Instructions for Students**

1. Upload the dataset

**Dataset:Sales_dataset** into your Google Colab / Jupyter Notebook.

2. Import the Pandas library and load the dataset.
3. Write Python code to solve each question.
4. Display both code and output for every question.
5. Use appropriate Pandas functions such as filtering, grouping, reshaping, and aggregation.

**Example to load the dataset:**

```
import pandas as pd

df = pd.read_csv("Pandas_advanced_sales_dataset.csv")

df.head()
```

**Q1 Dataset Exploration**

Perform the following exploratory steps:

- Display the first 10 rows
- Display the dataset structure using **info()**
- Find the number of rows and columns
- Identify the data types of each column

**Q2 Sales Filtering Analysis**

Filter the dataset to display:

1. Orders where Sales > 3500
2. Orders where Region = "East" and Sales > 2500

Explain what insights you observe.

**Q3 Customer Purchase Analysis**

Identify the top 5 customers based on total sales.

Hint: Use GroupBy with aggregation.

**Q4 Regional Sales Performance**

Calculate the total sales generated by each region.

Then identify:

- Which region generated the highest revenue?

**Q5 Product Performance Analysis**

- Calculate the total quantity sold for each product.
- Which product appears to be the most popular?

**Q6 Multi-Level Grouping**

- Group the dataset by: Region and Category

- Calculate the total sales for each combination.
- What insights can you derive from this analysis?

**Q7 Pivot Table Creation**

Create a pivot table showing:

- Rows → Region
- Columns → Product
- Values → Total Sales

Which region sells the most laptops?

**Q8 Profitability Analysis**

- Calculate the average profit for each product category.
- Which category is most profitable?

**Q9 Discount Impact Study**

Analyze the relationship between discount and profit.

Tasks:

- Calculate the average profit for each discount level
- Identify whether higher discounts lead to lower profit

**Q10 End-to-End Data Wrangling Task**

Perform the following workflow:

1. Filter the dataset where Sales > 3000
2. Group data by Region and Product
3. Calculate total sales
4. Reshape the result into a pivot table

Explain the business insights obtained from this analysis.